Earth’s temperature. More specifically, he estimated that if levels of carbon dioxide increased by 2.5 to 3 times, Arctic regions of the world would see temperature increases of 14–16°F (8–9°C). Arrhenius was building on the work of scientists Joseph Fourier and John Tyndall earlier in the 19th century. Fourier had wondered why Earth was not a freezing wasteland, when the Sun was too far away to heat it to its current temperature. He knew that heated surfaces—such as the surface of Earth—emit thermal energy, and that the thermal energy radiating back into space should result in colder temperatures on Earth. Something was regulating the temperature, and Fourier theorized that Earth’s atmosphere, made up of various gases, acted like a glass box, containing the air and keeping it warm. Fourier’s hypothesis, although oversimplistic, led to the “greenhouse effect” theory of Earth’s thermal regulation.

John Tyndall was the first to prove Fourier’s greenhouse effect hypothesis. His experiments demonstrated how, when Earth cools down at night—by releasing the heat absorbed from the Sun during the day—atmospheric gases, especially water vapor, absorb the heat (radiation) and cause a greenhouse effect. This keeps Earth’s temperature at an average 59°F (15°C), although in recent decades human activities that release greenhouse gases have pushed this figure higher. For example, the 10 warmest years on record have occurred since 2014, with 2024 being the warmest.